

THE GARDEN AT NIGHT

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CONCEPT

The Garden at Night is an interactive installation simulating a bioluminescent ecosystem: a night garden of glowing flowers, drifting moths, rising fireflies, and falling rain. Bioluminescent organisms glow not for beauty but for survival: to attract, to warn, to communicate. This piece takes that as its premise. The garden responds to the physical world around it (light levels, touch, atmospheric pressure translated through a dial), and the viewer becomes part of its environment rather than a spectator of it. Covering the light sensor collapses the garden into night. Turning the dial raises the wind. Pressing the button calls the rain and drives the moths to the edges of the scene.

INTERACTION DESIGN

Three physical inputs map to three environmental forces. Each has a corresponding LED on the circuit board that mirrors its effect on the screen, giving the operator immediate physical feedback without looking at the display.

- Potentiometer: wind strength. Bends stems, pushes moths, tilts fireflies. LED dims to bright.
- LDR sensor: time of day. Shifts sky colour, fog, flower colour, and star visibility. LED dims to bright.
- Button: rain toggle. Triggers drops, splashes, sky darkening, moth flee behaviour. LED stays on while active.

The session opens with a timed intro sequence. The hardware guide fades in before the intro clears, so the visitor understands the controls before the garden is revealed. Connecting the Arduino unlocks an arrow button that collapses the guide to a small, persistent strip, leaving the full canvas visible for the installation.

TECHNICAL STRUCTURE

The browser side runs in p5.js 1.7.0. Rendering uses a mix of p5 drawing primitives and the raw Canvas 2D API for operations p5 does not support, such as per-vertex gradients, radial fills, shadow compositing, and matrix shear transforms for wind-leaning trees. The Arduino sends a comma-separated line (pot, button, ldr) at 9600 baud every 30 ms. The browser reads this via the Web Serial API and maps the three values to wind strength, time of day, and rain intensity in real time. The codebase is split across seven files by concern: sketch.js (setup, draw, serial), helpers.js (wind, colour lerp, easing), background.js (sky, trees, moon, sun), environment.js (fog, rain, plants, stones), flowers_draw.js (stem, petals, bloom), creatures.js (moths, fireflies), and gardenSensors.ino (sensors, LEDs, serial output).